Architecture Development Part 2: Installation,Configuration, and Security of Openemr

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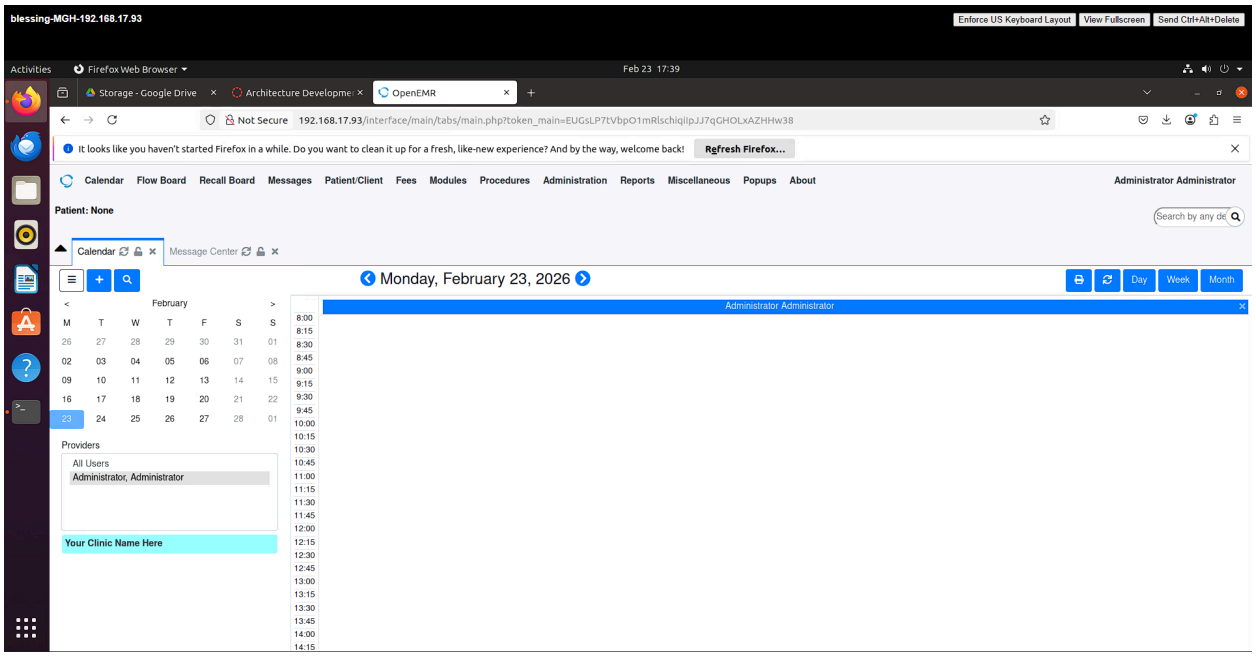

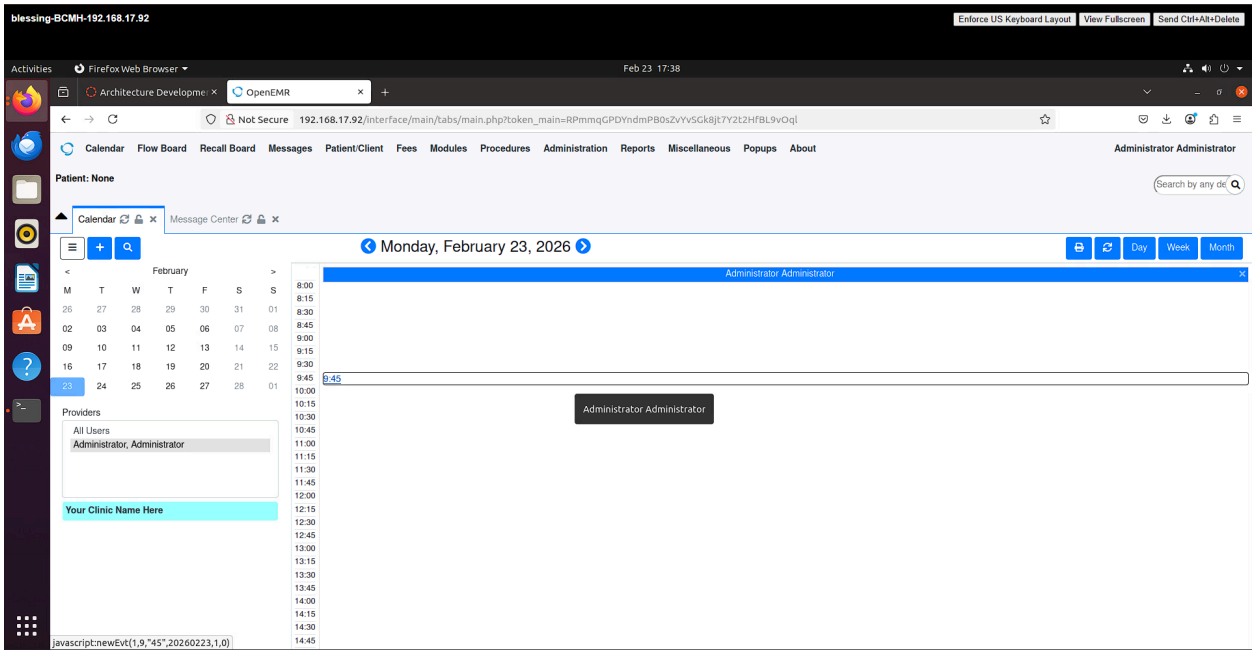
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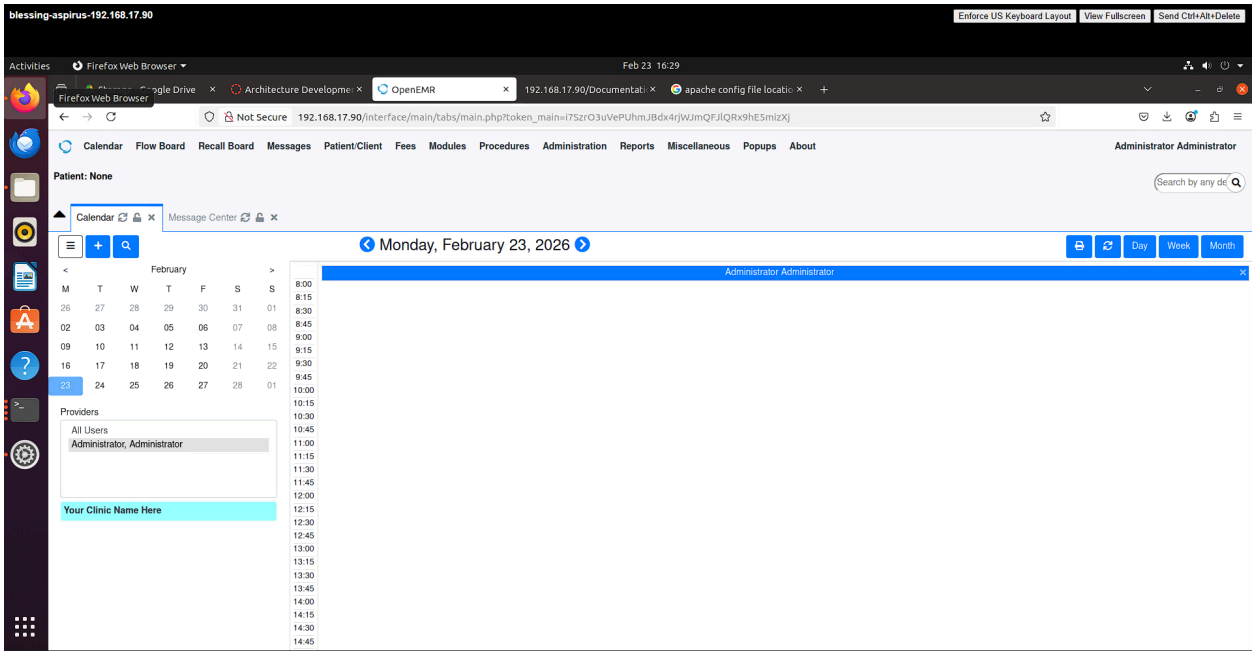
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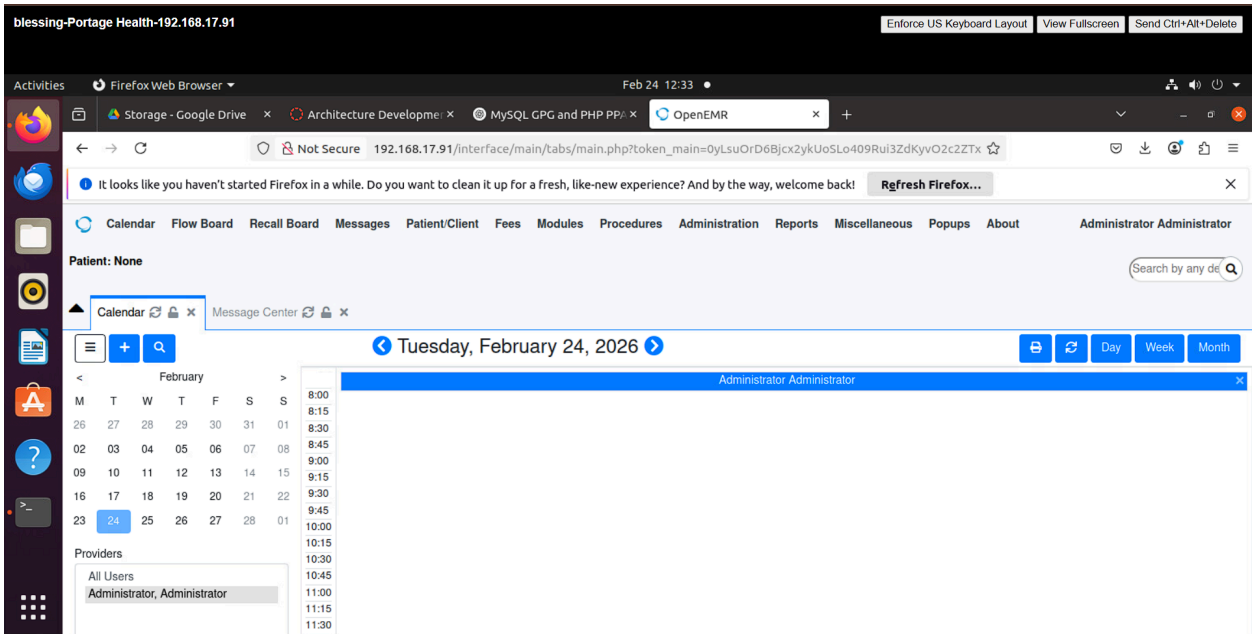


B STEPS TO SECURE OpenEMR

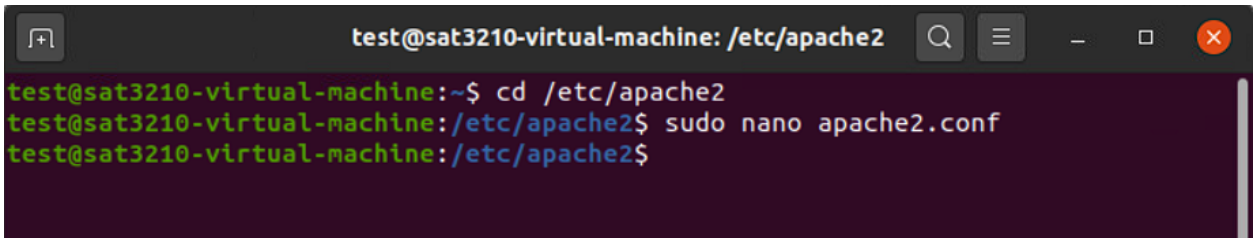
To secure OpenEMR,I used the following commands:
- sudo mysql_secure_installation
-sudo mysql -u root -p

To secure OpenEMR, I added the following configuration to the apache configuration file.

```
<Directory "/var/www/html/openemr">
    AllowOverride FileInfo
    Require all granted
 </Directory>
 <Directory "/var/www/html/openemr/sites">
    AllowOverride None
 </Directory>
<Directory "/var/www/html/openemr/sites/*/documents">
    Require all denied
 </Directory>
```

To locate the config file, I used the following steps:

1. Located the file directory using **cd /etc/apache2/**
2. Then I opened the config file using **sudo nano apache2.conf**



3. Then I added the above configuration at the bottom of the file, saved and exited.

```
test@sat3210-virtual-machine: /etc/php/7.4/apache2
  GNU nano 4.8                  /etc/apache2/apache2.conf
# see README.Debian for details.

# Include generic snippets of statements
IncludeOptional conf-enabled/*.conf

# Include the virtual host configurations:
IncludeOptional sites-enabled/*.conf

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
<Directory "/var/www/html/openemr">
      AllowOverride FileInfo
      Require all granted
  </Directory>
  <Directory "/var/www/html/openemr/sites">
      AllowOverride None
  </Directory>
  <Directory "/var/www/html/openemr/sites/*/documents">
      Require all denied
  </Directory>

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos
^X Exit       ^R Read File  ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line
```

C.Types of attacks that would still be susceptible to the OpenEMR platform:

-the openemr code is ‘HTTP’ and not ‘HTTPS’ and this shows that data is not encrypted even though there are login credentials

-session hijacking since OpenEMR uses session cookies after login and without HTTPS cookies are sent in plaintext and attackers can reuse them to impersonate admin